



United International University (UIU)
Dept. of Computer Science & Engineering (CSE)
Mid-Term Exam Trimester: Summer 2025

Course Code: CSE 4325 Course Title: Microprocessors and Microcontrollers
 Total Marks: 30 Duration: 1 hour 30 minute(s)

Any examinee found adopting unfair means would be expelled from the trimester/ program as per UIU disciplinary rules.

Question 1: Answer all the questions. (6 Marks)

Transfer of bus control from processor to device takes **200 ns**. Transfer of bus control from device to processor takes **300 ns**.

a.	One of the I/O devices employs DMA in cycle stealing mode and takes 148,456,500 ns to transfer a total 3648 Bytes of data. The I/O device has a data transfer rate of 1440 KB/minute . How many bytes are transferred at a time in this mode?	[3]
b.	In a Microcontroller, a seven segment display (like the image below) is connected to PORTC. The A, B, C, D, E, F and G pins are connected to PC2, PC1, PC5, PC7, PC4, PC0 and PC6 respectively. PC3 is not connected to anything. If you want to display the digit “6”, what should be the value of the register DDRC and why?	[2+1]

Question 2: Answer all the questions. (6 Marks)

a.	AX = 3A53 H, BX = 4D31 H, CX = 2BA1 H, Determine the values of the PF and OF after executing each of the following instructions sequentially. Provide a brief explanation for your answer. i. ADD AX, BX ii. SUB AX, CX	[4]
b.	What is meant by multiplexed pins ? Give an example of such pins with reference to the 8086 microprocessor.	[2]

Question 3: Answer all the questions. (6 Marks)

a.	CS = 1A32 H, DS = 6D2A H, SS = 7CA2 H SP = 3344 H, IP = 2211 H, SI = 11AA H I) To access the physical address in the Stack Segment, what should be the value of the segment register if the offset register holds the value of B294H.	[3]
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	II) To access the physical address in the Data Segment, what should be the value of the offset register if the segment register holds the value of 2B41H. Justify whether your answer is valid or not.	
b.	A DRAM has a data bus and an address bus that are twice the width of the data bus and address bus of an SRAM. The capacities of the SRAM and DRAM are 8 KB and 64 MB, respectively. Determine the width of the address bus of the SRAM and the width of the data bus of the DRAM.	[3]
<u>Question 4: Answer all the questions. (6 Marks)</u>		
a.	You need to design a system where a microprocessor with a 20-bit address bus and 16-bit data bus is interfaced to a 10 KB RAM system using full decoding, starting from address ACC00H. Each RAM chip has a 10-bit address bus and a 16-bit data bus. Draw a block diagram showing the microprocessor, RAM chips, and full decoding logic. Ensure the first RAM starts at ACC00H. Provide the 3rd, 5th, and 7th <i>last addresses</i> of the memory space occupied by the 2nd and 4th RAM chips.	[4+2]
<u>Question 5: Answer all the questions. (6 Marks)</u>		
a.	If the size of the interrupt vector table is 2Kbyte in an 8086 microprocessor, what will be the address of the lower byte of CS for the last interrupt number?	[3]
b.	An 8086 processor is executing an instruction at physical address 1C203H. IP = 2003H, Flag Register = F202H. Describe the exact sequence of events the 8086 performs when it encounters a software interrupt based on the given values and show the stack contents.	[3]